

2520090191 N .ROHITH

PRACTICAL WEEK 5

WEEK -5 Implement a producer-consumer communication system using anonymous pipes where the parent process generates data and the child process consumes it. Measure the communication efficiency.

Code:

```
nm PIDO 5.89 File: 51.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <time.h>
#include <string.h>

#define NUM_MESSAGES 10000
#define BUFFER_SIZE 100

int main()
{
    int pipefd[2];
    pid_t pid;
    char buffer[BUFFER_SIZE];

    struct timespec start, end;
    double elapsed_time;
    int i;

    /* Create anonymous pipe */
    if (pipe(pipefd) == -1)
    {
        perror("pipe");
        return 1;
    }

    /* Create child process */
    pid = fork();

    if (pid < 0)
    {
        perror("fork");
        return 1;
    }

    /* Parent Process - Producer */
    if (pid > 0)
    {
        close(pipefd[0]); // Close unused read end

        clock_gettime(CLOCK_MONOTONIC, &start);

        for (i = 0; i < NUM_MESSAGES; i++)
        {
            snprintf(buffer, BUFFER_SIZE, "Data %d", i + 1);

            if (write(pipefd[1], buffer, strlen(buffer) + 1) == -1)
            {
                perror("write");
                close(pipefd[1]);
                return 1;
            }
        }

        close(pipefd[1]); // Signal end of data

        wait(NULL); // Wait for child
    }
}
```

```
nrk@nallas-mac-Book-Air ~ % nano 51.c
nrk@nallas-mac-Book-Air ~ % gcc 51.c -o 51
nrk@nallas-mac-Book-Air ~ % ./51
Consumer completed successfully.

Producer completed successfully.
Number of messages: 10000
Communication time: 9.988604 seconds
Communication efficiency: 4151543.87 messages/second
nrk@nallas-mac-Book-Air ~ %
```

2. Develop a program that executes the equivalent of the shell command: `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

Code:

```
UW PIC0 8.09 File: 52.c Modified

/* Child 2: Execute grep ".c" */
close(pipefd[1]); // Close unused write end

/* Redirect stdin from pipe */
if (dup2(pipefd[0], STDIN_FILENO) == -1)
{
    perror("dup2");
    exit(1);
}

close(pipefd[0]);

/* Execute grep ".c" */
execlp("grep", "grep", ".c", (char *)NULL);

/* If execlp fails */
perror("execlp grep");
exit(1);
}

/* Parent process */
close(pipefd[0]);
close(pipefd[1]);

/* Wait for both child processes */
waitpid(pid1, NULL, 0);
waitpid(pid2, NULL, 0);

printf("\nCommand executed successfully.\n");

return 0;
}

Get Help Exit WriteOut Justify Read File Where is Prev Pg Next Pg Cut Text UnCut Text Cut Pos To Spell
```

Output:

```
nrk@nallas-mac-Book-Air ~ % gcc 52.c -o 52
nrk@nallas-mac-Book-Air ~ % ./52
1.c.save
11.c
14.c
15.c
16.c
17.c
19.c
2.c
20.c
21.c
22.c
23.c
24.c
25.c
26.c
27.c
3.c
30.c
4.c
444.c
5.c
50.c
51.c
52.c
a.c.save
ancient.dress.fbx
Applications
b.c.save
DaVinci Resolve Media
ddca.circ
Documents
fed.css.h
file.c
fl.c
kl.c
lk.c
logism.circ
Music
n.c
package.com.mediflow.java
Pictures
project.html
Public
public class SegmentTreeLazy {.java
r.c
rohith.c
space.save
Untitled-1.cs

Command executed successfully.
nrk@nallas-mac-Book-Air ~ %
```

